

“Still Changing After All These Years”: A Dynamic Analysis of Portuguese Local Government Service Delivery Choices (1999-2025)

António Tavares * & Pedro Camões ±

* Research Center in Political Science (CICP), School of Economics, Management and Political Science, University of Minho, Portugal

± Research Unit on Governance, Competitiveness and Public Policies (GOVCOPP), University of Aveiro, Portugal

Abstract

This paper revisits Portuguese municipal corporatization as a dynamic delegation choice rather than a one-off shift from bureaucracy to corporation. Using a municipality-service-cohort panel for 278 mainland municipalities from 1999 to 2025, organized into nine three-year cohorts, it examines when local governments use municipally owned corporations, when they enter or exit corporatized provision, and how the 2012 legal-fiscal tightening reshaped those choices. The evidence shows that corporatization expanded through the late 2000s, narrowed after reform, and became increasingly concentrated in infrastructure and utility-like services. Fixed-effects status, entry, and exit models support a revised political transaction-cost account. Asset specificity is the most robust service predictor, while measurability operates less consistently than standard monitoring arguments imply. The strongest regularity is own-service persistence: once a service is corporatized, it tends to remain so, whereas cross-service spillovers are weak. Fiscal conditions matter more than partisan control, and exits are largely selective rather than random.

1 Introduction

Across advanced democracies, municipalities increasingly corporatize public service delivery by creating municipally owned corporations, local public enterprises, and other company forms under public ownership and political control. This shift has been widely observed, but it has not produced a single “municipal corporation model.” Instead, municipal corporations vary in legal form, financing, scope, and governance arrangements, and they sit uneasily between the administrative logics of the bureaucracy and the corporate logics of enterprise organization (Cepiku, Mussari & Giordano, 2016; Skelcher, 2017; Voorn, Van Genugten, & Van Thiel, 2017; Grossi & Thomasson, 2023). Fragmented evidence suggests that municipalities corporatize some services but not others, and the same municipality may corporatize at one point and later terminate companies, merge them, or reintegrate activities into the bureaucracy when political control, fiscal constraints, or regulatory frameworks change (Tavares & Camões, 2007; Camões & Rodrigues, 2021; Tavares & Camões, 2023; Maine, 2025).

A large comparative literature explains local organizational choices using fiscal stress and intergovernmental dependency, ideology and partisan control, economies of scale, administrative capacity, and transaction and monitoring problems (Brown & Potoski, 2003; Bel & Fageda, 2017; Andrews, Ferry, Skelcher, & Węgorowski, 2020; Krause and Van Thiel, 2019). Yet much of this evidence base remains either cross-sectional, focused on a single service, or limited to short windows that cannot fully capture the dynamics of corporatization, particularly the possibility of reversal and the interdependence of choices across municipal service portfolios (Tavares & Camões, 2007; Voorn et al., 2017; Grossi & Thomasson, 2023; Tavares & Camões, 2023). That empirical gap matters because corporatization is rarely a one-off institutional event. Municipalities govern multiple services simultaneously, and organizational investments made to establish and supervise municipal companies in one domain can spill over into others through common legal templates, accounting routines, managerial skill, and political “know-how” about how to make corporatization governable (Schoute, Budding, & Gradus, 2017; Moldenæs & Torsteinsen, 2017; Klausen & Winsvold, 2019).

This paper starts from the simple premise that municipal corporatization is a repeated, portfolio-wide governance choice. Treating each corporatization decision as a separate one risks mischaracterizing both the determinants and dynamics when decision moments are connected within municipalities across services and over time. Recent work on interdependent policy choices makes the broader point that institutional decisions often travel in bundles, and that estimators assuming independence can yield misleading inferences when organizational choices are mutually reinforcing (Peixoto et al., 2024). In the corporatization context, this motivates a second-order question that has been asserted more often than it has been tested: does corporatizing one service increase the probability of corporatizing other services in subsequent years, through learning, common templates, and the accumulation of political-administrative capacity for delegation and oversight (Krause and Van Thiel, 2019; Andrews et al., 2020; Voorn et al., 2017)?

We develop and test a dynamic political transaction costs (PTC) account of corporatization. Building on transaction cost reasoning in the public sector, this account emphasizes that elected officials face political costs when delegating service delivery, particularly the costs of specifying what is to be done, monitoring performance, managing blame, and sustaining control over organizations that are structurally more autonomous than bureaucratic units (Tavares & Camões, 2007; Brown & Potoski, 2003; Andrews et al., 2020). Service measurability and asset specificity remain central because they shape both the feasibility of monitoring and the political risks of opportunism, underperformance, and contested blame attribution (Tavares & Camões, 2007; Brown & Potoski, 2003; Bel & Fageda, 2017). Our first theoretical and empirical task, therefore, is longitudinal: do these classic PTC determinants continue to structure corporatization when we exploit within-municipality variation over time rather than relying primarily on cross-sectional differences?

As corporatization diffuses and normalizes, as municipalities accumulate experience, and as governance investments become sunk, the calculus of the PTC itself may shift. This opens the door to a mechanism of path dependence, which is often invoked but rarely tested jointly in corporatization research. Once municipalities invest in corporate governance arrangements, professional networks, and political oversight routines, the perceived costs of exit rise, potentially producing persistence even when conditions change (Pierson, 2000; Voorn et al., 2017; Tavares & Camões, 2023).

Finally, corporatization unfolds within temporal contexts that can reweight political and administrative constraints. Fiscal crises, austerity, and emergency governance can change the feasibility and desirability of corporatized provision by altering intergovernmental oversight, tightening financial rules, shifting salience, and intensifying blame risks (Ferry et al., 2018; Camões & Rodrigues, 2021; Maine, 2025). In Portugal, changes in the legal framework governing the local corporate sector and shifts in fiscal oversight are likely to have altered not only adoption incentives but also the prospects for termination and reintegration (Camões & Rodrigues, 2021; Tavares & Camões, 2023). This suggests a third question: how do shocks—especially austerity and the COVID period—interact with fiscal dependency, party control, and service characteristics to shape both entries into and exits from corporatized provision?

Empirically, we study corporatization as a binary choice, i.e., corporatization versus in-house bureaucracy, using a complete municipal panel for the 278 municipalities in mainland Portugal from 1999 to 2025, allowing us to observe adoption and reversal over time (Camões & Rodrigues, 2021; Tavares & Camões, 2023). This design aligns corporatization research with the broader insight from public management that governance arrangements are dynamic and can move out and back in, even when the institutional form is corporatization rather than contracting-out (Hefetz & Warner, 2004; Tavares & Camões, 2023; Maine, 2025). By explicitly modeling multi-service interdependence, entry, and exit, we aim to move from a determinants-focused literature to an account of corporatization trajectories that connects service characteristics, political transaction costs, and time-varying constraints in a single analytic framework.

The paper makes four contributions. First, it provides a dynamic validation and refinement of a PTC framework by testing service-level and political–institutional determinants of corporatization in a long panel rather than a single cross-section. Second, it theorizes and tests cross-service contagion through learning-by-doing and governance spillovers that reduce the fixed costs of establishing and controlling additional municipal companies. Third, it treats external shocks as conditioning environments that reshape PTC and help explain both adoption and reversal under stress. Finally, it provides a methodological template for analyzing interdependent longitudinal governance choices with multiple correlated binary outcomes, a design intended to travel to other national contexts with substantial local corporate sectors.

2 Corporatization as a delegation decision in municipal service delivery

This paper conceptualizes municipal corporatization primarily as a delegation decision in local public service delivery. A municipality decides that a given service will be provided by an organizationally distinct, municipally controlled entity with separate legal personality, rather than remaining within the municipal bureaucracy. The emphasis is not on an organizational boundary moving, but on elected officials choosing to delegate operational responsibility to an arm’s-length public organization while retaining ultimate public authority. Framing corporatization in this way enhances the generalizability of our findings, despite substantial cross-national variation in legal labels, governance rules, and the broader menu of delivery modes available to local governments (Cepiku et al., 2016; Skelcher, 2017; Voorn et al., 2017).

2.1. At arm’s-length public ownership with delegated management

We treat corporatization as the creation or use of a municipally controlled organization that has separate legal status, holds distinct governance and managerial arrangements relative to the municipal administration (e.g., executive board and management and internal procedures not embedded in the municipal chain of command), and remains ultimately subject to public authority through ownership rights, appointment powers, fiscal ties, and statutory oversight (Tavares and Camões, 2010). This definition emphasizes delegated service provision under public ownership rather than legal form alone. Corporatization is neither outright privatization nor merely a procurement contract; it is a public-sector hybrid in which politicians delegate operational authority to an organizationally separate entity, potentially increasing managerial flexibility while altering the structure of political control and accountability (Voorn et al., 2017; Andrews et al., 2020; Klausen and Winsvold, 2019).

This delegation decision is analytically important because it reconfigures decision rights, monitoring and reporting relations, the visibility of costs and performance, and the political allocation of credit and blame (Krause and Van Thiel, 2019; Klausen and Winsvold, 2019). These are precisely the domains in which transaction-cost arguments about governance structures have been extended to the public sector, including the claim that service attributes such as asset specificity and measurability condition the attractiveness of alternative production arrangements (Brown and Potoski, 2003; Tavares and Camões, 2007; Bel and Fageda, 2017). What must be

explained, therefore, is why local politicians choose to delegate service provision to an arm's-length public corporation in some cases while keeping the same service inside the municipal hierarchy in others.

2.2. In-house bureaucratic production

The comparison category is in-house production through the municipal administrative apparatus. In-house provision implies that production and management are conducted within the municipal hierarchy and under the prevailing public-law administrative constraints on budgeting, personnel, procurement rules, and operations. Relative to corporatization, in-house provision typically entails less delegation of operational authority, greater direct political and administrative supervision, and a clearer integration of managerial decisions into ordinary municipal accountability channels (budgetary control, administrative reporting, and political scrutiny) (Voorn et al., 2017; Krause and Van Thiel, 2019; Klausen and Winsvold, 2019).

The article, therefore, adopts the deliberately binary empirical focus of corporatization versus in-house, because it isolates a common delegation choice across many countries: whether municipalities provide services through their own bureaucracy or through at arm's-length public entities that they own and control (Cepiku et al., 2016; Andrews et al., 2020). This focus enables the analysis of municipal corporatization as a distinct mode of delegated public provision without requiring a jurisdiction-specific enumeration of every feasible delivery arrangement.

2.3. Why a general theory of delegated corporatization

The article keeps the choice set conceptual. The theory is about the relative attractiveness of in-house hierarchical provision versus delegated service delivery through corporatized, at arm's length public ownership. Cross-national differences are treated as scope conditions and parameter shifts, not as additional "choices" to be modeled. In particular, institutional rules governing corporatized entities, such as appointment and dismissal rights, audit and disclosure requirements, procurement constraints, debt consolidation and fiscal oversight, and the capacity of central and local watchdog institutions, are expected to alter the magnitude of PTC without changing the core logic that corporatization delegates operational authority and reshapes monitoring and accountability (Klausen and Winsvold, 2019; Krause and Van Thiel, 2019; Grossi and Thomasson, 2023).

This conceptualization implies four mechanisms that organize the theoretical framework and the hypotheses. The first mechanism connects service attributes and governance suitability. Consistent with public-sector transaction cost arguments, services vary in the ease with which outputs can be specified and evaluated (measurability) and the extent to which effective delivery depends on non-redeployable investments (asset specificity) (Williamson, 1981). These features shape monitoring costs, vulnerability to opportunism, and the relative attractiveness of keeping provision inside the local hierarchy versus delegating it to an arm's-length corporation (Brown and Potoski, 2003; Tavares and Camões, 2007; Bel and Fageda, 2017).

Second, political–institutional context affects the management of delegation. Corporatization substitutes one set of constraints for another. It can relax administrative rigidities yet increase political agency and accountability risks because delegated providers are harder to direct through ordinary bureaucratic command (Voorn et al., 2017; Krause and Van Thiel, 2019; Moldenæs and Torsteinsen, 2017). The fiscal position and intergovernmental dependence of municipalities, their bureaucratic capacity, and the degree of political fragmentation condition both the demand for corporatization and the ability to control it. Related work in Portugal and elsewhere highlights how wealth, fiscal conditions, and political incentives can be pivotal when deciding over externalization options, including municipal corporations (Tavares and Camões, 2007; Andrews et al., 2020).

Third, because municipal service portfolios are diverse, corporatization is never a one-shot decision. Once municipalities acquire the legal, managerial, and political capabilities required to delegate services to corporatized entities and govern them effectively, subsequent corporatization decisions may become easier, and experiences in one domain may spill over into others, producing interdependence within a municipality’s service portfolio (Schoute, Budding, and Gradus, 2018; Moldenæs and Torsteinsen, 2017; Klausen and Winsvold, 2019). This emphasis on interdependent institutional choices aligns with recent evidence that service delivery decisions may be jointly structured rather than independent across decision moments (Peixoto et al., 2024), even if our empirical focus here is strictly corporatization versus in-house.

Lastly, environmental shocks may act as re-adjustment devices. Major shocks (fiscal crises, pandemics) can reshape the balance between flexibility, control, and accountability, altering the costs of maintaining in-house capacity and the political risks of delegated delivery through municipal corporations (Ferry et al., 2018; Camões and Rodrigues, 2021; Maine, 2025). A central claim of this paper is that shocks should be treated as conditioning environments for corporatization dynamics, rather than merely as time dichotomous variables.

3 Political Transaction Costs in Dynamic and Multi-service Portfolios

Following Tavares and Camões (2007, 2010), we treat corporatization as a political governance-structure choice in which municipalities compare the costs of producing a service inside the bureaucracy with the costs of delegating that service to an arm’s-length public corporation. The basic intuition derives from the classic works of Oliver Williamson, i.e., the idea that governance forms differ because transactions differ, and municipalities choose the arrangement that minimizes the sum of production and governance costs rather than any single cost category in isolation (Williamson, 1979; Ferris and Graddy, 1986, 1991). In the municipal setting, however, governance costs are political as well as administrative. In-house bureaucracies can be rigid, slow, and costly to adapt because personnel, budgeting, and operating decisions remain embedded in municipal hierarchy and public-law constraints. Corporatization may relax some of those internal bureaucratic costs, but it introduces a diverse set of delegation costs, including specifying objectives, monitoring managers, maintaining political control, and managing accountability when provision is moved to a legally separate body (Tavares and Camões, 2007, 2010; Brown and Potoski, 2003; Overman, 2016; Bel and Fageda, 2017).

The original argument of Tavares and Camões (2007) remains the starting point. Corporatization should be more attractive where the expected savings from escaping bureaucratic constraints exceed the added PTC of delegation. Service characteristics matter because they shape those delegation costs directly. When outputs are relatively measurable and the service does not depend heavily on highly specific, non-redeployable assets, delegation is easier to structure and monitor. In contrast, when outputs are opaque and investments are highly specific, municipalities have stronger reasons to retain direct hierarchical control (Tavares and Camões, 2007, 2010; Brown and Potoski, 2003; Hefetz and Warner, 2004; Bel and Fageda, 2017). Municipal political and fiscal conditions matter because they shift costs: fiscal dependence, administrative capacity, and partisan incentives alter the relative weight of bureaucratic costs and delegation costs, but they do so through this same underlying trade-off (Ferris and Graddy, 2001; Tavares and Camões, 2007; Andrews et al., 2020).

The main extension in this paper is dynamic. Once a municipality creates a corporation, the governance technology of delegation changes for that municipality. Legal templates, accounting routines, oversight procedures, board-appointment practices, and political expectations become partly sunk. These investments lower the marginal cost of keeping an existing corporation in place and may also lower the cost of creating another one later. Put differently, corporatization can generate its own capabilities, routines, and reputational commitments, making subsequent decisions path dependent rather than fresh comparisons between two untouched alternatives (Andrews et al., 2020; Voorn et al., 2017; Krause and Van Thiel, 2019; Tavares and Camões, 2023). This is also why reversal matters analytically. Termination is not simply the absence of corporatization, but the active discarding of an existing delegated arrangement, often after legal, political, and managerial commitments have already accumulated (Camões and Rodrigues, 2021).

A second extension is portfolio logic. Municipalities do not face a single, once-and-for-all choice about service delivery; they govern a portfolio of services simultaneously, and each service in that portfolio demands its own decision about how it should be delivered. Across this portfolio, local governments must choose, and periodically revise, among alternative arrangements, including direct in-house provision, arm's-length municipal corporations, inter-municipal cooperation, and contracting out to private or non-profit operators. Thus, the experience of running one corporation in a sector can spill over into other domains by providing legal and financial know-how, managerial routines, and political learning about how to control an arm's-length provider. In that sense, corporatization in one service may reduce the perceived uncertainty and organizational fixed costs of corporatization in another service. The broader literature on local public organizations and collaborative arrangements points in this direction by arguing that governance capability is often cumulative, and arrangements become easier to reproduce once municipalities learn how to operate them (Schoute, Budding, and Gradus, 2017; Moldenæs and Torsteinsen, 2017; Klausen and Winsvold, 2019; Aldag and Warner, 2018; Peixoto et al., 2024).

Finally, the trade-off itself can be reweighted by shocks. Fiscal crises, austerity, and emergency conditions can change municipal constraints, intensify blame risks, and alter the attractiveness of flexibility relative to direct control. On one hand, shocks may push municipalities toward delegation

because they need flexibility or distance from politically costly operations. On the other hand, they may encourage recentralization because control, transparency, and fiscal oversight become more valuable when municipalities face external shocks. For this reason, shocks are better treated as conditioning environments than as a source of a single directional hypothesis. They are expected to affect the probability of both entry into and exit from corporatization by changing how municipalities value bureaucratic costs and delegation costs at a given moment (Ferry et al., 2018; Camões and Rodrigues, 2021; Maine, 2025).

The observable implication is straightforward. If PTC remain the correct baseline theory, then service characteristics should still structure corporatization choices. If the dynamic extension is correct, those same choices should also display persistence over time and spillovers across services within municipalities. Thus, in order to keep the theory tight, we formulate only the hypotheses for which the extended PTC argument yields a clear directional expectation. Municipal fiscal and political variables are treated as cost-shifting covariates in the empirical analysis, but they do not each receive stand-alone hypotheses. The hypotheses focus instead on the baseline service trade-off and both dynamic extensions: time and portfolio logic.

Hypothesis 1: Services with more measurable outputs and lower asset specificity are more likely to be corporatized than services with less measurable outputs and higher asset specificity, because delegation costs are lower relative to the costs of in-house bureaucratic provision.

Hypothesis 2: Holding service and municipal characteristics constant, prior corporatization of a service increases the probability that the same service remains corporatized in subsequent periods. Existing corporatization creates sunk governance investments, routines, and switching costs that make reversal less likely and persistence more likely.

Hypothesis 3: Holding service and municipal characteristics constant, corporatization in one service increases the probability of corporatization in other services within the same municipality. Experience with one delegated arrangement lowers the marginal organizational and political costs of delegating additional services.

4 Empirical setting

4.1 Why Portugal is a valuable test

Portugal is a particularly valuable setting for studying the dynamics of municipal corporatization because it combines a clear legal form, unusually broad territorial coverage, and a long-term horizon that spans both expansion and retrenchment in the number of municipal corporations. Since the late 1990s, Portuguese municipalities have been allowed to create municipal and intermunicipal corporations for a wide range of local functions, making corporatization a real and recurrent governance option rather than an exceptional organizational arrangement. At the same time, the Portuguese case is analytically sound because while municipalities vary sharply in size

and fiscal capacity, they operate within a common national legal framework and share a common portfolio of services. This combination allows studying repeated organizational choices under shared institutional rules while observing substantial local variation.

The period covered in this paper is also unusually informative for a dynamic analysis. The early phase of the sector was comparatively open and flexible, as evidenced by the fact that municipal corporations expanded in number and scope, and corporatization was often seen as a practical way to escape some bureaucratic rigidities of in-house administration while retaining public ownership and political control. This produced meaningful variation not only in whether municipalities corporatized, but also in which services they corporatized and how broad their portfolios became. Portugal, therefore, provides a setting in which entry, persistence, restructuring, merger, and dissolution can all be observed within the same institutional context.

The key turning point is the post-bailout legal reform of 2012, especially Law 50/2012, which fundamentally altered the environment in which municipal corporations operated. The reform replaced a relatively permissive framework with a more restrictive, closely supervised one by tightening the legal criteria for creation and persistence, strengthening fiscal and economic viability requirements, and increasing pressure to restructure or dissolve corporations that no longer met the new standards. In substantive terms, this matters because the paper is not studying corporatization in a stable institutional environment. It is studying the same organizational choice under two very different regimes: an earlier phase in which corporatization was comparatively available and flexible, and a later phase in which the sector became more constrained, more scrutinized, and more politically costly to maintain. This makes Portugal not only a useful case for testing the baseline PTC argument, but also an unusually strong setting for examining how legal and fiscal shocks reshape the trajectory of municipal corporatization over time.

4.2 Data collection

Our dataset combines two types of sources: pre-existing raw files assembled by the authors over many years and a second layer of data collected specifically for this project. The first set includes General Directorate of Local Authorities (DGAL) spreadsheets and reports, legacy research files and student-collected spreadsheets, Banco de Portugal extracts, and municipality-level demographic, fiscal, and electoral files. The second family consists of reproducibly generated datasets that harmonize names, dates, ownership, sectors, municipalities, and service areas across those heterogeneous sources. This strategy was necessary because no single official source provides a complete annual history of municipal and intermunicipal corporations from the late 1990s to the mid-2020s.

On the corporation side, the core raw materials are DGAL snapshot files for 2010, 2017, 2021, and 2025, DGAL monitoring annexes, annual Court of Auditors spreadsheets, and a small archive of early official documents collected from Official Gazette (*Diário da República*) and related public repositories. These sources contained different kinds of information: some identify whether a corporation existed in a given year, others provide financial figures, and others help recover dates of company incorporation, dissolution dates, municipality affiliation, employment, websites, and

operating sector. DGAL registry entries and corporate webpages were also harvested through an AI-assisted, script-based workflow to capture registry fields, website links, and activity descriptions in a reproducible way. Because corporation names are sometimes inconsistent across files, all entities were linked through a normalized identifier and then assigned a readable canonical name, producing a list of companies and two related corporation-year panels: a stricter observed panel and a broader historically extended panel.

On the municipality side, we assembled a municipality-year panel for mainland Portugal by integrating the existing fiscal, demographic, and institutional files in a large archive, which provides standardized municipal indicators over time, and adding the electoral results from the Ministry of Internal Administration for municipal executive elections. Municipality names were harmonized across sources, ambiguous cases were resolved through override files and National Institute of Statistics (INE) codes, and intermunicipal corporations were expanded to all owner municipalities when the analysis required municipality-level attribution. This step was critical for building a consistent municipality x year backbone to which corporation information could be merged.

The analytical unit for the main models is the municipality x service x time cell. Each corporation was classified into one of seven substantively interpretable service areas: culture, science and events; economic development, markets, street fairs and business parks; environment, water, sewerage and waste; public space, municipal facilities and maintenance; sports, recreation, leisure and tourism; social housing; and transport, mobility and parking. The classification was generated with a transparent codebook using firm names, sector codes, website descriptions, and fallback rules, so that the service taxonomy remains close to the Tavares and Camões (2007) framework while being better aligned with the evidence available for the present study.

From these harmonized source files we constructed three kinds of variables. First, service-level variables identify whether a municipality had at least one active corporation in a given service and whether that service entered into or exited from corporatized provision. Second, municipal-level variables capture political control, fiscal structure, population size, and demographic composition. Third, period variables capture changing institutional environments, including the post-2012 legal framework, austerity, and the COVID period. The raw annual panel covers mainland municipalities from 1999 to 2025, but because coverage is uneven in some years, the paper's main analyses are presented in nine three-year cohorts, which smooth short breaks without abandoning the temporal logic of the argument.

A final distinction concerns two versions of the corporatization series. The strict observed series is the conservative one: a municipality-service-year is coded as corporatized only when that observation is directly supported by the strongest documentary sources. The extended historical series is broader: when yearly observation is incomplete, it uses confirmed historical linkage and lifecycle information to reconstruct longer runs of service presence. Intuitively, the strict series tells us what we can observe directly with greater confidence, while the extended series tells us what the longer trajectory of the service most plausibly looks like once gaps in annual observation are taken into account. We retain both because they serve different purposes. The strict series is the

more cautious benchmark for precise observation, whereas the extended series is more useful for long-run dynamics, entries, exits, and sequence analysis. All transformation steps were scripted and documented in the project's code and methodological notes so that the dataset can be rebuilt and audited rather than manually edited.

5 Evolution of Municipal Corporatization Choices

5.1 Service mix changes over time

This section uses the cohort-based panel to describe how municipal corporatization evolved in the 1999-2025 period. This descriptive evidence serves two purposes. First, it shows whether corporatization expanded or contracted over time and whether that evolution was concentrated in particular services or spread across broader municipal portfolios. Second, it provides a first sense of the temporal breaks that matter for interpretation, especially the expansion up to the late 2000s, the post-2012 adjustment, and the uneven reconfiguration of the sector in the later period.

The overall pattern is not one of simple linear diffusion. Under the extended historical series, the number of mainland municipalities with at least one corporatized service rises from 105 in 1999-2001 to 143 in 2008-2010, remains high in 2011-2016, falls back to 119 in 2017-2019, and then stabilizes around the low 140s in the two most recent cohorts. What changes more substantially is the internal composition of corporatization: the sector becomes broader up to the mid-2010s and then more selectively concentrated afterward.

Figure 1 - Evolution of Service Prevalence

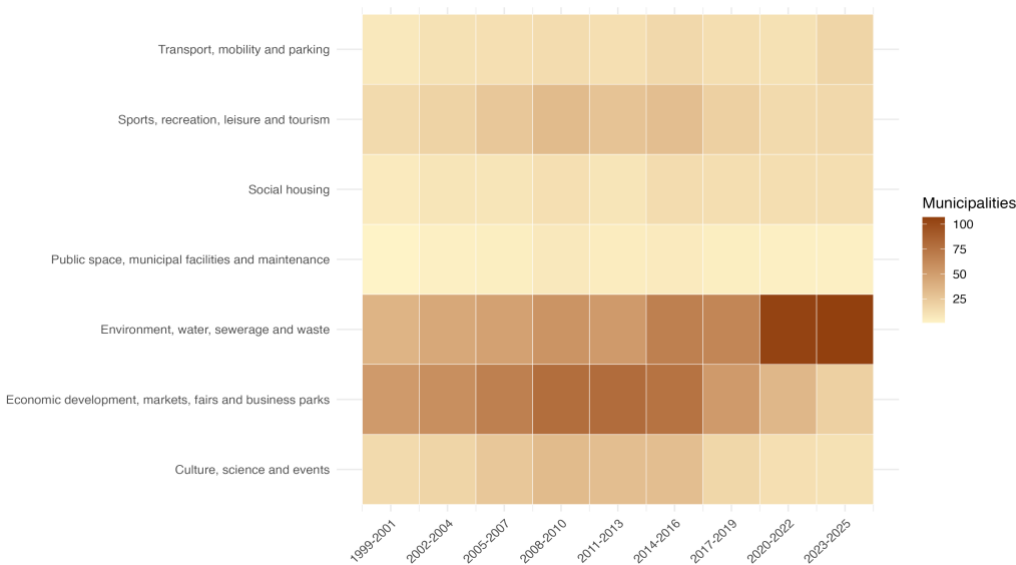
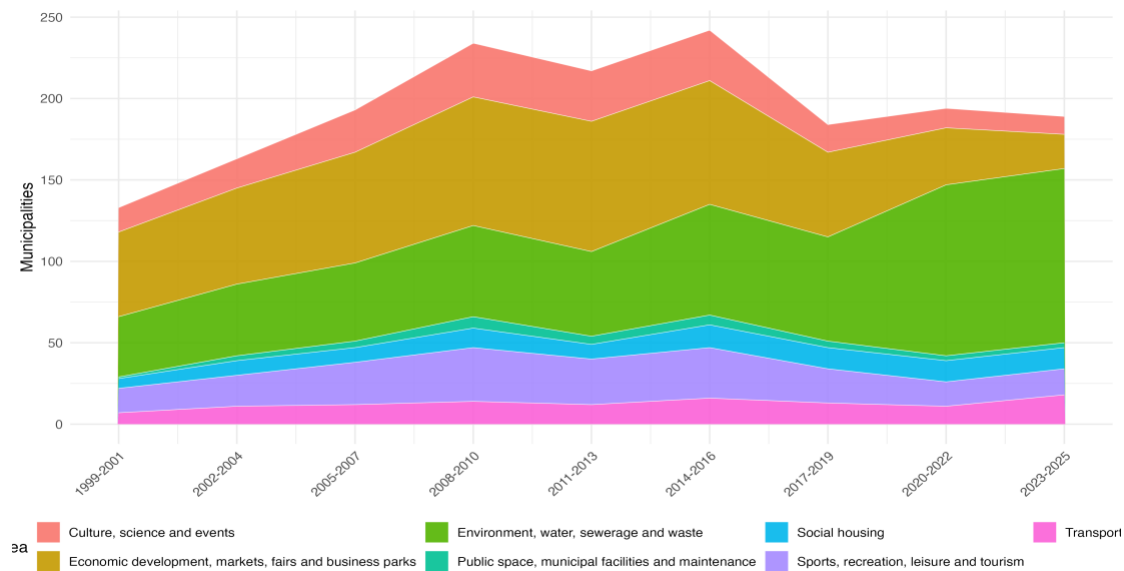


Figure 1 provides a compact heatmap of service prevalence across cohorts, while Figure 2 presents the same evolution in a format that makes changes in magnitude easier to read. Read together, the

two figures suggest that the Portuguese local corporate sector changed not only in size, but also in functional composition. The clearest descriptive finding is that early corporatization was led by economic-development activities and environmental utilities, but the long-run balance between these two areas changed sharply over time. In the first 3-year cohort, *economic development, markets, street fairs, and business parks* was the most prevalent service area, appearing in 52 municipalities, compared with 37 for environment, water, sewerage, and waste. Economic-development corporatization continued to expand through 2011-2013, reaching 80 municipalities, but then contracted markedly to 21 by 2023-2025. Environmental services followed a different trajectory: after growing more gradually in the early period, they became the dominant corporatized field in the later cohorts, reaching 105 municipalities in 2020-2022 and 107 in 2023-2025.

Figure 2 - Service Mix over time



Other service areas remained smaller and more cyclical. *Culture, science, and events* and *sports, recreation, leisure, and tourism* both expanded up to 2008-2010, when each appeared in 33 municipalities, but both receded afterward and remained well below those levels in the final cohorts. Social housing stayed comparatively modest but stable, moving from 6 municipalities in 1999-2001 to 13 in 2023-2025. Transport, mobility, and parking remained a secondary but persistent field, while public space, municipal facilities, and maintenance never became a large domain of corporatized provision. The descriptive implication is that municipal corporatization did not spread evenly across local functions: it became increasingly concentrated in services with stronger infrastructural and utility characteristics, while some of the more discretionary developmental and recreational domains lost ground after the reform period.

5.2 Size of service portfolio

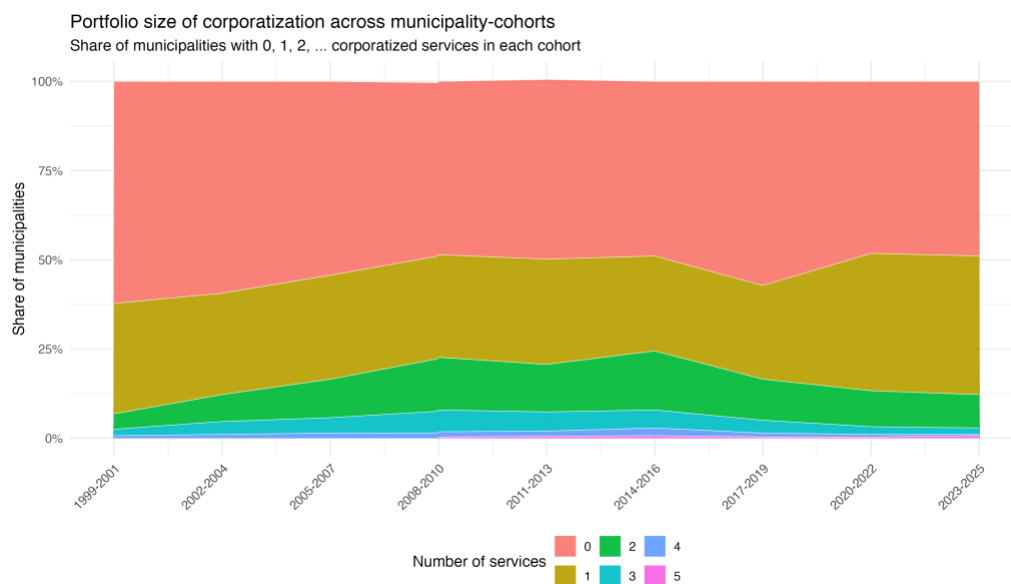
Portfolio size shows whether corporatization is mostly a single-service choice or a broader organizational strategy. Up to the late 2000s, corporatization clearly became more widespread and

more multi-service. The share of municipalities with no corporatized services falls from 62.2 percent in 1999-2001 to 48.6 percent in 2008-2010, while the proportion with two or more corporatized services rises from less than 7 percent to nearly 23 percent over the same period. This suggests that expansion was not driven only by first adoptions. Many municipalities were also broadening their corporatized portfolios.

That broadening continued into 2014-2016, when almost one quarter of municipalities had at least two corporatized service areas and a small number had four or five. After that point, however, the portfolio pattern narrowed. By 2023-2025, almost half of municipalities again had no corporatized services, and the most common positive outcome was not a broad portfolio but a single corporatized service. In other words, the later configuration of the sector is not simply smaller than the expansionary phase; it is also more concentrated. Municipal corporatization becomes less a strategy of accumulating multiple enterprises across service domains and more a selective choice focused on a narrower set of functions.

Figure 3 makes this shift visible by contrasting the earlier rise of multi-service portfolios with the later predominance of zero- and one-service municipalities. Descriptively, the figure supports a portfolio interpretation of corporatization: municipalities learned to use the corporate form across multiple domains during the expansionary period, but that broader organizational model became harder to sustain under the more restrictive environment later.

Figure 1 - Portfolio size over time

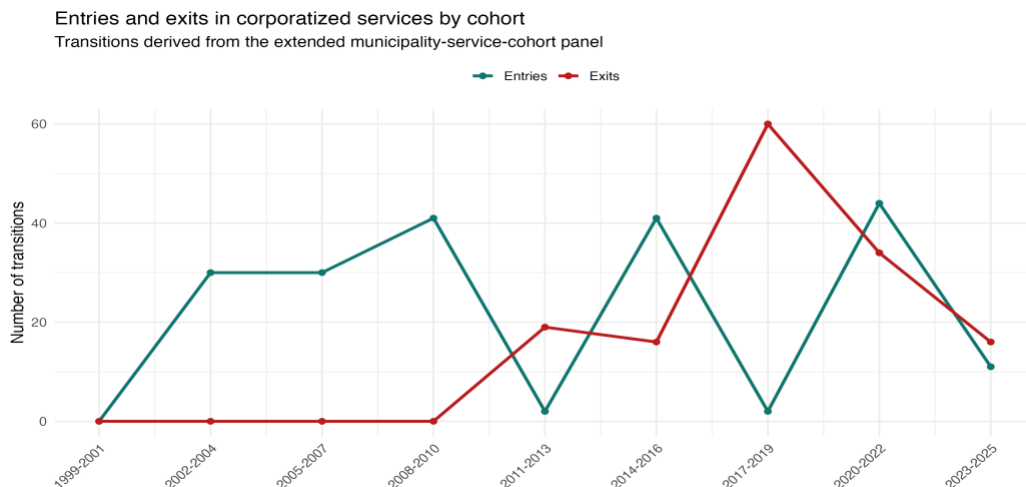


5.3 Transition dynamics (entries and exits)

A dynamic analysis should pay close attention to transitions, not just levels, because entries and exits tell us whether change is driven by cumulative expansion, by retrenchment, or by both. Here the cohort structure is especially useful. Entries dominate the first four cohorts, rising from 30 in 2002-2004 and 2005-2007 to 41 in 2008-2010, while exits are essentially absent from the early expansionary phase. The first major break appears in 2011-2013, when entries collapse to 2 and exits rise to 19, coinciding with the period in which the sector moved from a permissive to a much more restrictive legal and fiscal environment.

The second major break comes in 2017-2019, when only 2 entries are recorded against 60 exits. This is the clearest episode of contraction in the panel. The subsequent cohort, 2020-2022, shows renewed movement in both directions, with 44 entries and 34 exits, suggesting partial reconfiguration rather than simple recovery. In the final cohort, transitions moderate again, with 11 entries and 16 exits. Taken together, Figure 4 shows that corporatization in Portugal is neither a one-way diffusion process nor a single shock story. It is a path-dependent but reversible organizational trajectory, marked by long periods of stability interrupted by sharper waves of entry and exit.

Figure 2 - Entries and Exits



This interpretation is reinforced by the overall transition profile of the cohort panel. Across all adjacent cohort comparisons, most municipality-service cells remain in the same state, with nearly 89 percent staying non-corporatized and just over 9 percent staying corporatized, while entries and exits are relatively rare events at the cell level. The descriptive section therefore points to a double conclusion that motivates the empirical models below: corporatization is highly inertial once service-by-service trajectories are taken seriously, but when change does occur it is clustered in identifiable periods and unevenly distributed across service domains and municipal portfolios.

6 Determinants of Choice

6.1 *Estimation strategy*

The regression analysis is organized around the municipality x service x cohort panel. This is the most appropriate unit for the paper because the theoretical question is not simply whether a municipality has municipal corporations in general, but whether it corporatizes a given service and how that decision evolves over time. The design therefore follows the service-delivery choice directly, while preserving the dynamic dimension that is central to the paper's argument.

The empirical strategy is best understood as a dynamic panel framework with strong fixed effects rather than as a causal identification design in the narrow sense. Municipality or municipality-service fixed effects absorb stable differences across local governments and service domains, while cohort fixed effects absorb common temporal shocks such as macroeconomic conditions, legal changes, and broader sectoral trends. The models should therefore be read as theory-guided association models that exploit within-panel variation, not as designs that can isolate causal effects from nationally shared institutional changes.

This framework supports several types of comparison at once: across services within municipalities, across municipalities facing different fiscal and political contexts, across periods before and after major institutional change, and across different stages of the corporatization process, including status, persistence, entry, and exit.

The empirical section uses four related model types. Status models estimate whether a municipality-service-cohort cell is corporatized. Persistence models keep the same dependent variable but make dynamic continuity explicit by testing whether prior corporatization strongly predicts current corporatization. Entry models focus on transitions from non-corporatized to corporatized service provision, while exit models focus on the reverse transition from corporatized to non-corporatized provision.

The status models are estimated using fixed-effects logistic regression. The persistence, entry, and exit specifications are estimated as linear fixed-effects models. This combination keeps the binary service-choice logic at the center of the analysis while making the interpretation of transition dynamics more transparent. It also implies that coefficients in the status models should be interpreted as changes in log-odds and not directly compared numerically with the linear coefficients reported for persistence, entry, and exit.

The paper's main regression narrative is limited to the choice models. Complementary count models based on the number of corporatized service areas in each municipality are retained only as robustness checks in the appendix, because they address the scale of municipal corporatization portfolios rather than the paper's primary question of service-by-service governance choice. That appendix is also useful for interpreting variables such as population, because it allows a contrast between within-municipality dynamic estimates and between-municipality portfolio comparisons.

6.2 Variables

6.2.1 Dependent Variables

The analysis uses three main dependent variables. *corporatized* equals 1 when a municipality has at least one corporation active in a given service during the cohort. *entry* equals 1 when a municipality-service cell moves from non-corporatized in the previous cohort to corporatized in the current cohort. *exit* equals 1 when a municipality-service cell moves from corporatized in the previous cohort to non-corporatized in the current cohort.

In the persistence specifications, the lagged value of *corporatized* is introduced as a key explanatory variable in order to capture institutional continuity. This allows the models to distinguish the general correlates of corporatization from the narrower question of whether, once established, corporatized arrangements tend to endure.

6.2.2 Main Explanatory Variables

The first explanatory block captures structural characteristics of services. *measurability* records how easily service outputs and performance can be observed and monitored, with higher values indicating greater observability. *asset_specificity* captures the degree to which service provision depends on specialized or place-bound assets that are not easily redeployed. *user_fee_potential* indicates whether the service can plausibly support direct charging to users. These variables are attached to the service area rather than to individual municipality-cohort observations and operationalize the transaction-cost logic developed in earlier literature.

The second block captures the dynamic institutional context. *lag_corporatized* tests whether corporatization is path-dependent, while *lag_other_services_corporatized* captures lagged cross-service portfolio spillovers, that is, whether prior corporatization in other service domains makes subsequent corporatization of the focal service more likely.

The third block introduces the municipal political, fiscal, and demographic context. In the cleaned specifications, the political block is represented by *left_mayor* and *majority_share*, the fiscal block by *fiscal_dependency* and *own_tax_share*, and the socioeconomic block by *log_population* and *elderly_share*. Substantively, this block allows the paper to examine whether the service-delivery choice is shaped only by transaction-cost properties or also by the broader political and fiscal environment in which municipalities govern, while avoiding unnecessary overlap among closely related covariates.

6.2.3 Time Interactions

Because the service-characteristic variables are structural and theory-driven, the paper also estimates a second layer of models in which those traits are interacted with the post-2012 legal environment. The purpose is not to replace the baseline service logic, but to test whether its empirical relevance changes after the institutional restructuring of the municipal corporate sector that followed the bailout period and Law 50/2012.

The following table provides compact descriptive statistics for the main outcome and explanatory variables used in the empirical models.

Table 1 - Descriptive Statistics

Variable	Mean	SD	Min	Max	N
Corporatized service	0.100	0.300	0.000	1.000	17514
Entry into corporatization	0.013	0.113	0.000	1.000	15568
Exit from corporatization	0.009	0.096	0.000	1.000	15568
Lagged corporatization	0.100	0.300	0.000	1.000	15568
Other corporatized services	0.599	0.835	0.000	5.000	17514
Service measurability	2.429	0.728	1.000	3.000	17514
Asset specificity	1.571	0.495	1.000	2.000	17514
User-fee potential	0.714	0.452	0.000	1.000	17514
Fiscal dependency	0.386	0.142	0.044	0.797	17514
Own-tax share	0.197	0.147	0.014	0.729	17514
Capital-revenue share	0.274	0.136	0.007	0.642	17514
Majority share	0.612	0.107	0.286	1.000	17514
Winner vote share	51.591	8.426	26.830	83.120	17514
Left mayor	0.503	0.471	0.000	1.000	17514
Right mayor	0.425	0.479	0.000	1.000	17514
Log population	9.783	1.120	7.260	13.256	17514
Elderly share	0.248	0.071	0.084	0.471	17514
Youth share	0.238	0.044	0.119	0.396	17514
Purchasing power index	3.603	11.562	0.053	203.853	17493

6.3 Findings

The regression results are organized by dependent variable. Table 2 reports the status models, including both baseline and persistence specifications because they share the same binary outcome. Table 3 displays the entry models, whereas Table 4 reports the exit models. This organization keeps the empirical narrative centered on the three main substantive questions of the paper: whether a service is corporatized, whether corporatization is adopted, and whether corporatization is abandoned.

Table 2 - Results for the Status Models

	Baseline Model	Post-2012 Model	Persistence Model	Persistence Post-2012	Political Model	Fiscal and Social Model	Context Model
Measurability	-0.257*** (0.046)	-0.613*** (0.070)					
Asset specificity	0.952*** (0.067)	0.545*** (0.101)					
User-fee potential	-0.072 (0.077)	-0.155 (0.113)					
Log population	-1.335** (0.487)	-1.382** (0.490)	-0.100*** (0.030)	-0.100*** (0.030)		-4.175** (1.520)	-4.388** (1.533)
Fiscal dependency	0.202 (0.665)	0.207 (0.669)	0.014 (0.029)	0.014 (0.029)		0.186 (1.780)	-0.188 (1.801)
Majority share	-0.415 (0.386)	-0.423 (0.388)	-0.025+ (0.015)	-0.025+ (0.015)	-1.191 (0.976)		-1.429 (0.997)
Lag corporatization			0.591*** (0.017)	0.585*** (0.016)	3.692*** (0.202)	3.665*** (0.204)	3.660*** (0.205)
Lagged other corpor. services			-0.003 (0.003)	-0.002 (0.003)	-0.094 (0.180)	0.016 (0.180)	-0.001 (0.182)
Measurability x post-2012		0.653*** (0.099)		0.044*** (0.005)			
Asset specificity x post-2012		0.738*** (0.143)		0.037*** (0.007)			
User-fee potential x post-2012		0.150 (0.163)		0.016* (0.007)			
Left mayor					-0.298 (0.236)		-0.299 (0.239)
Own-tax share						-4.194+ (2.547)	-4.311+ (2.546)
Elderly share						-9.636 (7.194)	-10.276 (7.266)
N	11844	11844	15568	15568	1912	1912	1912
AIC	8737.18	8652.01	-15430.07	-15674.17	2018.31	2011.06	2010.85
BIC	10227.85	10164.82	-445.55	-666.69	3407.29	3411.15	3422.05
R-squared	0.160	0.169	0.818	0.821	0.427	0.431	0.433

Table 2 provides the clearest evidence that the service-level transaction-cost logic remains relevant over time, but it also shows that the strongest empirical regularity in the Portuguese case is persistence. In the two baseline status specifications, measurability is negative and statistically significant, while asset specificity is positive and statistically significant. The coefficients are sizeable in the fixed-effects logit models: the coefficient on measurability ranges from about -0.26 to -0.61, whereas the coefficient on asset specificity ranges from about 0.55 to 0.95, both with conventional levels of statistical significance. User fee potential, by contrast, is not statistically distinguishable from zero in the baseline specifications.

Substantively, the status models suggest that service characteristics still matter, but their interpretation must be read dynamically rather than statically. The positive effect of asset specificity indicates that corporatization remains particularly associated with services requiring

dedicated, infrastructure-heavy, or place-bound investments. The negative coefficient on measurability, taken on its own, would suggest that corporatization was initially more concentrated in less easily monitored service areas. However, this baseline coefficient should not be read in isolation, because the interaction model shows that the relationship changes over time. In that sense, the more informative interpretation is not simply that less measurable services are corporatized, but that the earlier pattern is progressively reweighted as the institutional context changes.

The interaction specification adds an important temporal refinement. The interactions *measurability x post-2012* and *asset_specificity x post-2012* are both positive and highly significant, meaning that the service profile of corporatization changes after the 2012 legal reform. For measurability, the implication is that the earlier negative association weakens sharply and, in practice, moves toward a neutral or slightly positive relationship in the later period. Put more intuitively, as time passes and especially after the reform, corporatization becomes relatively more concentrated in services that are easier to monitor. For asset specificity, the positive interaction means that the later period reinforces even more strongly the concentration of corporatization in services with specific, place-bound, and harder-to-redeploy assets. The interaction for *user_fee_potential x 2012* is positive but not statistically significant. In substantive terms, the post-2012 legal environment did not eliminate the importance of service characteristics; rather, it made corporatization more selective and increasingly concentrated in services that were both easier to monitor and harder to reintegrate once organized through arm's-length entities.

The persistence columns show that the dominant empirical feature of the Portuguese sector is institutional continuity. *lag_corporatized* is large, positive, and precisely estimated in both persistence models, while the lagged cross-service spillover variable is small and not statistically significant. Substantively, this means that once a municipality-service cell is corporatized it is very likely to remain corporatized in the next cohort, but prior corporatization elsewhere in the municipal portfolio does not add much explanatory power once own-service persistence is taken into account. Thus, the Portuguese pattern looks much more like service-specific path dependence than cumulative diffusion across broad municipal portfolios.

The political, fiscal-social, and combined context models refine that picture further. In these specifications, the political variables remain weak: neither *left_mayor* nor *majority_share* reaches conventional significance levels in the fuller status models. The fiscal block is also more modest than in the earlier, more crowded specifications. *own_tax_share* remains the most informative element, but its effect is only suggestive rather than strongly robust once overlapping covariates are removed. The safest interpretation, therefore, is not that the context variables provide a single powerful explanation of corporatized status, but that they help show which dimensions remain more plausible than others once the institutional and service-specific structure is taken into account. The clearest contextual result is the negative coefficient on *log_population*. This coefficient should be interpreted carefully. In these fixed-effects choice models, it is not a comparison between larger and smaller municipalities. Rather, it is a dynamic within-municipality estimate: it asks whether, holding constant stable municipal differences and common cohort

shocks, changes in population within the same municipality over time are associated with changes in the probability that a given service is corporatized. Read in this way, the negative sign does not contradict the broader literature showing that larger municipalities often have more municipal corporations in cross-sectional terms. It instead suggests that, within this panel and for this service-level choice, population growth within the same municipality over time is associated with a lower probability that a given service will be corporatized. The complementary count models reported in the technical appendix make this distinction explicit by showing that the between-municipality count relationship can differ from the within-municipality dynamic one.

Table 3 - Entry Models

	Baseline Model	Post-2012 Model	Political Model	Fiscal and Social Model	Context Model
Log population	0.024 (0.020)	0.020 (0.020)		0.024 (0.021)	0.024 (0.021)
Fiscal dependency	0.045+ (0.023)	0.046* (0.023)		0.046+ (0.023)	0.045+ (0.023)
Majority share	-0.004 (0.010)	-0.003 (0.010)	-0.007 (0.010)		-0.004 (0.010)
Lagged other corporatized services	0.001 (0.003)	0.001 (0.003)	0.001 (0.002)	0.001 (0.003)	0.001 (0.003)
Measurability x post-2012		0.025*** (0.003)			
Asset specificity x post-2012		0.031*** (0.005)			
User-fee potential x post-2012		0.009* (0.005)			
Left mayor			0.002 (0.004)		0.002 (0.004)
Own-tax share				-0.008 (0.029)	-0.009 (0.029)
Elderly share				-0.006 (0.067)	-0.009 (0.067)
N	13972	13972	13972	13972	13972
AIC	-21663.21	-21801.96	-21660.26	-21661.17	-21657.75
BIC	-7743.04	-7859.15	-7747.63	-7733.45	-7714.94
R-squared	0.266	0.274	0.266	0.266	0.266

Table 3 shifts the focus from status to adoption, and the results are noticeably more selective. In the baseline entry model, few variables are consistently strong. The lagged cross-service spillover variable is small and not statistically significant, *majority_share* is near zero, and *log_population* is positive but statistically weak. The only variable approaching conventional significance is *fiscal_dependency*, which remains positive and marginally significant. Technically, this indicates that entry is harder to explain than status: adoption is a relatively rare event, and the coefficients are correspondingly smaller and less stable.

The interaction specification nevertheless reveals an important pattern. All three post-2012 interaction terms are positive, and both *measurability x post-2012* and *asset_specificity x post-2012* are strongly significant, while *user_fee_potential x post-2012* is also statistically significant. Substantively, this suggests that services enter corporatization after the 2012 reform more selectively than before. In a more restrictive institutional environment, new corporatization appears increasingly concentrated in service domains whose outputs are easier to measure, whose assets are more specific, and whose financing is at least partly compatible with direct charging. This is also broadly consistent with one of the reform's likely goals: not necessarily to eliminate corporatization altogether, but to narrow it to activities that could be more easily justified, monitored, and disciplined under a tighter legal and fiscal framework. We return to this point in the discussion, because the post-2012 pattern seems to reflect not only contraction, but also institutional filtering.

The contrast models again show the limited role of political indicators. The political-only specification produces no meaningful effects for *left_mayor* or *majority_share*. In the fiscal-social and combined context models, *fiscal_dependency* remains the most important predictor, while the broader fiscal pattern is less uniformly strong. The substantive interpretation is that entries into corporatization remain more closely related to municipal fiscal capacity than to partisan control, but that this effect is narrower and more selective once overlapping fiscal and demographic covariates are trimmed away.

Table 3, therefore, reinforces two broader points. First, the determinants of entry are not identical to the determinants of status. A municipality may sustain corporatized provision for long periods even when the conditions associated with new adoption are weak. Second, the post-2012 environment seems to have made entry more selective and more tightly conditioned by both service characteristics and municipal fiscal position.

Table 4 - Exit Models

	Baseline Model	Post-2012 Model	Political Model	Fiscal and Social Model	Context Model
Log population	0.191 (0.172)	0.039 (0.146)		0.158 (0.175)	0.179 (0.171)
Fiscal dependency	0.503** (0.174)	0.464** (0.166)		0.475** (0.173)	0.514** (0.176)
Majority share	0.133 (0.103)	0.141 (0.102)	0.089 (0.103)		0.132 (0.102)
Lagged other corporatized services	0.002 (0.023)	0.004 (0.023)	0.002 (0.022)	-0.001 (0.023)	0.003 (0.022)
Measurability x post-2012		-0.064 (0.043)			
Asset specificity x post-2012		-0.231*** (0.044)			
User-fee potential x post-2012		-0.129+ (0.076)			
Left mayor			0.043 (0.029)		0.044 (0.029)

	Baseline Model	Post-2012 Model	Political Model	Fiscal and Social Model	Context Model
Own-tax share				0.205 (0.244)	0.196 (0.244)
Elderly share				-0.283 (0.881)	-0.293 (0.859)
N	1511	1511	1511	1511	1511
AIC	573.15	507.98	577.72	576.00	574.23
BIC	2009.70	1960.48	2008.94	2017.86	2026.73
R-squared	0.288	0.321	0.285	0.288	0.290

Table 4 addresses the abandonment of corporatized provision. These are the most difficult models in the paper, and that difficulty should be stated explicitly. Exit is a rarer and more institutionally specific event than status or entry, so the specifications are naturally less stable. Even so, several useful patterns emerge.

In the baseline exit model, *fiscal_dependency* is positive and statistically significant, suggesting that higher dependence on intergovernmental transfers is associated with a greater likelihood of abandoning corporatized service provision. This pattern weakens in the broader context models, where the coefficient loses significance once additional fiscal and sociodemographic controls are added. Technically, this means that the fiscal story behind exit is less robust than the one observed for status and entry. Substantively, however, the baseline result remains consistent with the idea that more dependent municipalities may face tighter constraints when maintaining at arm's-length organizations, which become politically or administratively costly.

The post-2012 interaction model adds the clearest service-level result in the exit table. *asset_specificity x post-2012* is negative and highly significant, indicating that after the legal tightening of 2012, services with more specific and less redeployable assets became less likely to exit corporatization. *user_fee_potential x post-2012* is also negative, though only marginally significant, and *measurability x post-2012* remains statistically weak. The substantive implication is that once the post-2012 environment increased scrutiny and constraints, municipalities appear to have been more willing to terminate corporatization in services that were easier to reabsorb or reorganize, while retaining corporatized arrangements in services with stronger sunk infrastructure and more specialized operating requirements.

As in the other models, the political variables remain weak. Neither *left_mayor* nor *majority_share* produces stable effects in the exit specifications. The most robust contextual variable is *fiscal_dependency*, which is positive and statistically significant in the combined exit specification. The remaining controls are comparatively weak. These results should still be treated with caution, but they suggest that exit is more closely related to fiscal strain than to partisan control once the contextual block is simplified.

Overall, Table 4 supports a more qualified conclusion than the previous tables. Exit is not well explained by partisan politics, and it is less systematically structured than status or entry. However, the results do suggest that retrenchment is not random. It appears more common under

some forms of fiscal strain and less common in asset-specific services after the 2012 legal break. This is consistent with the broader descriptive argument of the paper: corporatization in Portugal is reversible, but reversals are selective rather than indiscriminate.

Taken together, the three tables support a coherent interpretation of corporatization as a dynamic but uneven governance choice. The strongest regularity is persistence: once a service is corporatized, it tends to remain corporatized. Around that inertial core, service characteristics and municipal context still matter, but the more parsimonious specifications sharpen the institutional conclusion. Asset specificity remains the most robust structural service predictor, and the fiscal variables generally outperform the political variables as potential explanations. Most importantly, the evidence now points much more strongly to own-service path dependence than to broad cross-service spillover. In the Portuguese case, corporatization appears to persist primarily because established service-specific arrangements endure, not because prior corporatization elsewhere in the municipal portfolio systematically triggers new corporatization across the board.

Equally important, the post-2012 interaction models indicate that the determinants of corporatization did not remain constant across the whole period. The legal and fiscal reordering of the local corporate sector did not simply reduce corporatization uniformly. Instead, it appears to have made both adoption and retention more selective, especially in relation to service characteristics. That finding fits the paper's broader claim that corporatization must be understood as an evolving governance choice embedded in changing institutional environments rather than as a one-off organizational decision.

7 Discussion

The Portuguese evidence suggests that the most portable element of the argument is not any particular legal form, but the underlying delegation problem. Wherever municipalities create legally separate, publicly owned organizations, elected officials face a familiar trade-off: corporatization may relax bureaucratic constraints, but it also raises the political costs of specification, monitoring, financial control, and blame management. In this sense, the PTC framework travels well. Service characteristics, fiscal constraints, and accumulated experience should matter in other countries too, even if their coefficients and substantive meanings are likely to vary across institutional settings.

What appears less portable is the exact pattern of corporatization. The positive association between asset specificity and corporatized status, the weakening role of measurability after reform, and the post-2012 selectivity of entry are best read through Portugal's legal and fiscal trajectory. The Law 50/2012 did not merely reduce the number of municipal corporations; it changed the conditions under which corporatization could be created, justified, maintained, or abandoned. Under tighter viability rules, stronger reporting requirements, and closer oversight, corporatization became less a broadly available escape from municipal bureaucracy and more a selective organizational form concentrated in services where sunk infrastructure, operating specialization,

and revenue possibilities made municipal reintegration costly or unattractive. These mechanisms should be expected in other countries only to the extent that their national frameworks impose comparable financial reporting, debt consolidation, audit, and dissolution pressures.

The dynamic evidence also refines PTC as a general framework. The results do not support a static version in which municipalities repeatedly compare in-house and corporatized provision from scratch. Instead, the strongest regularity is persistence. Once a municipality-service cell is corporatized, it is likely to remain so. This implies that PTC parameters change over time. Initial corporatization creates legal templates, accounting routines, managerial relationships, and political expectations that alter the later cost of maintaining or abandoning the arrangement. At the same time, the weak evidence for broad cross-service spillovers cautions against treating corporatization experience as a general municipal capability that automatically spreads across the whole portfolio of services. In Portugal, path dependence appears more service-specific than portfolio-wide.

This distinction matters theoretically. Normalization does not necessarily mean continuous diffusion. A corporatized form can become familiar while remaining harder to adopt, easier to scrutinize, and more selective in its use. The Portuguese case shows that corporatization may pass through phases: expansion under permissive rules, consolidation or retrenchment under fiscal stress, and selective persistence where exit costs are high. PTC should therefore be treated as a time-varying framework in which institutional reforms reweight the costs of autonomy, control, flexibility, and accountability.

The policy implication is that national corporatization frameworks shape local choices not only by allowing or forbidding municipal companies, but also by defining the conditions for delegation throughout the lifecycle. Entry rules, viability tests, accounting standards, debt treatment, transparency obligations, and dissolution procedures all affect the local calculus. A permissive framework may encourage broad experimentation but can also generate later correction costs. A restrictive framework may improve fiscal discipline and accountability, but it may also channel corporatization toward a narrower set of infrastructure-heavy or revenue-compatible services. The central design question is therefore not whether corporatization is good or bad, but whether the national framework aligns autonomy with credible oversight throughout creation, operation, restructuring, and exit.

8 Conclusion

This paper revisits municipal corporatization as a dynamic governance choice rather than a one-time decision to move services outside the bureaucracy. Using long-run evidence from Portuguese municipalities, it shows that political transaction costs remain a useful framework, but only if treated dynamically. Service characteristics still matter, especially after the post-2012 legal tightening, yet the strongest empirical regularity is persistence: once a service is corporatized, it tends to remain corporatized. Corporatization, therefore, changes the conditions of later choice by

creating routines, legal templates, oversight arrangements, and sunk political and organizational commitments.

The theoretical contribution is to refine the PTC framework by showing that its parameters are not fixed. They evolve as corporatization becomes normalized, as municipalities accumulate experience, and as national rules alter the costs of entry, maintenance, and exit. The paper also distinguishes service-specific path dependence from broader portfolio diffusion. In the Portuguese case, corporatization persists mainly within established service areas, while evidence of generalized spillovers across municipal portfolios is weaker.

The paper also offers a design template for studying corporatization elsewhere. Analytically, it treats corporatization as a repeated municipality-service choice, separates status from entry and exit, and exploits institutional breaks to examine how legal and fiscal environments reshape governance decisions. This approach can be applied to other countries with substantial local corporate sectors, provided that researchers account for national differences in legal forms, reporting requirements, financial consolidation, audit capacity, and dissolution rules.

For governance reform, the implication is that corporatization frameworks should regulate the full lifecycle of local public enterprises. Rules for creation matter, but so do viability tests, transparency obligations, oversight capacity, and credible procedures for restructuring or municipal reintegration of services previously under corporatized form. A permissive framework may encourage experimentation but also generate later correction costs. In contrast, a restrictive framework may strengthen accountability but narrow corporatization toward services where exit is costly or financing is clearer.

A future extension is the consideration of cross-municipal influences. Corporatization choices are embedded in a broader organizational field in which municipalities observe one another, borrow templates, and respond to what becomes viewed as normal or legitimate. Peer effects need not displace political transaction costs; they can work through them by changing beliefs about the feasibility, controllability, and legitimacy of delegation. If neighboring or comparable municipalities adopt corporatized provision, the perceived risks of doing the same may fall even when service characteristics are unchanged. This is the local government version of diffusion and institutional isomorphism, and it is consistent with work on corporatization, hybridity, and normalization in public-sector organizations (DiMaggio and Powell, 1983; Skelcher, 2017; Grossi and Thomasson, 2023; Maine, 2025). This hypothesis should be explored in future work.

References

Aldag, Austin M., & Warner, Mildred (2018). Cooperation, not cost savings: explaining duration of shared service agreements. *Local Government Studies*, 44(3), 350-370.
<https://doi.org/10.1080/03003930.2017.1411810>

Andrews, Rhys, Ferry, Laurence, Skelcher, Chris, & Wegorowski, Piotr (2020). Corporatization in the Public Sector: Explaining the Growth of Local Government Companies. *Public Administration Review*, 80(3), 482-493. <https://doi.org/10.1111/puar.13052>

Bel, Germà, & Fageda, Xavier (2017). What have we learned from the last three decades of empirical studies on factors driving local privatisation?. *Local Government Studies*, 43(4), 503-511. <https://doi.org/10.1080/03003930.2017.1303486>

Brown, Trevor L., & Potoski, Matthew (2003). Contract-Management Capacity in Municipal and County Governments. *Public Administration Review*, 63(2), 153-164. <https://doi.org/10.1111/1540-6210.00276>

Brown, Trevor L., & Potoski, Matthew (2003). Managing Contract Performance: A Transaction Costs Approach. *Journal of Policy Analysis and Management*, 22(2), 275-297. <https://doi.org/10.1002/pam.10117>

Camões, Pedro J., & Rodrigues, Miguel (2021). From enthusiasm to disenchantment: an analysis of the termination of Portuguese municipal enterprises. *Public Money and Management*, 41, 387-394. <https://doi.org/10.1080/09540962.2020.1763605>

Cepiku, Denita, Mussari, Ricardo, & Giordano, Fillipo (2016). Local governments managing austerity: Approaches, determinants and impact. *Public Administration*, 94(1), 223-243. <https://doi.org/10.1111/padm.12222>

DiMaggio, P. J., & Powell, W. W. (1983). The iron cage revisited: Institutional isomorphism and collective rationality in organizational fields. *American Sociological Review*, 48(2), 147-160. <https://doi.org/10.2307/2095101>

Ferris, James, & Graddy, Elizabeth (1986). Contracting out: For What? With Whom?. *Public Administration Review*, 46(4), 332-344. <https://doi.org/10.2307/976307>

Ferris, James M., & Graddy, Elizabeth (1991). Production costs, transaction costs, and local government contractor choice. *Economic Inquiry*, 29(3), 541-554. <https://doi.org/10.1111/j.1465-7295.1991.tb00845.x>

Ferry, Laurence, Andrews, Rhys, Skelcher, Chris, & Wegorowski, Piotr (2018). New development: Corporatization of local authorities in England in the wake of austerity 2010-2016. *Public Money & Management*, 38(6), 477-480. <https://doi.org/10.1080/09540962.2018.1486629>

Grossi, Giuseppe, & Thomasson, Anna (2023). New development: Are the holding companies as a hybrid governance model reinforcing the control on municipal corporations?. *Public Money & Management*, 43(1), 77-79. <https://doi.org/10.1080/09540962.2022.2101269>

Hefetz, Amir, & Warner, Mildred E. (2004). Privatization and its reverse: Explaining the dynamics of the government contracting process. *Journal of Public Administration Research and Theory*, 14, 171-190. <https://doi.org/10.1093/jopart/muh012>

Klausen, Jan Erling, & Winsvold, Marte (2021). Corporate governance and democratic accountability: local state-owned enterprises in Norway. *Journal of Public Policy*, 41(1), 161-184. <https://doi.org/10.1017/S0143814X19000229>

Krause, Tobias, & Van Thiel, Sandra (2019). Perceived managerial autonomy in municipally owned corporations: disentangling the impact of output control, process control, and policy-profession conflict. *Public Management Review*, 21(2), 187-211. <https://doi.org/10.1080/14719037.2018.1473472>

Maine, Joshua (2025). Municipal corporations in national and disciplinary contexts: A systematic literature review and future research agenda. *International Public Management Journal*, 28(2), 217-237. <https://doi.org/10.1080/10967494.2024.2438764>

Moldenæs, Turid, & Torsteinsen, Harald (2017). Re-politicisation as post-NPM response? Municipal companies in a Norwegian context. *Local Government Studies*, 43(4), 512-532. <https://doi.org/10.1080/03003930.2017.1305954>

Overman, Sjors (2016) Great expectations of public service delegation: A systematic review. *Public Management Review*, 18:8, 1238-1262. <https://doi.org/10.1080/14719037.2015.1103891>

Peixoto, Ânia, Camões, Pedro J., & Tavares, António (2024). Local delivery of bus transportation services: a model of interdependency of provision and production choices. *Journal of Economic Policy Reform*, 27(2), 185-208. <https://doi.org/10.1080/17487870.2024.2330438>

Pierson, Paul (2000). Increasing Returns, Path Dependence, and the Study of Politics. *American Political Science Review*, 94(2), 251-267. <https://doi.org/10.2307/2586011>

Rodrigues, Miguel, Tavares, Antonio F., & Araújo, J. Filipe (2012). Municipal Service Delivery: The Role of Transaction Costs in the Choice between Alternative Governance Mechanisms. *Local Government Studies*, 38(5), 615-638. <https://doi.org/10.1080/03003930.2012.666211>

Schoute, Martijn, Budding, Tjerk, & Gradus, Raymond (2018). Municipalities' Choices of Service Delivery Modes: The Influence of Service, Political, Governance, and Financial Characteristics. *International Public Management Journal*, 21(4), 502-532. <https://doi.org/10.1080/10967494.2017.1297337>

Skelcher, Chris (2017). An enterprising municipality? Municipalisation, corporatisation and the political economy of Birmingham City Council in the nineteenth and twenty-first centuries. *Local Government Studies*, 43(6), 927-945. <https://doi.org/10.1080/03003930.2017.1359163>

Tavares, António F., & Camões, Pedro J. (2007). Local service delivery choices in Portugal: A political transaction costs framework. *Local Government Studies*, 33(4), 535-553. <https://doi.org/10.1080/03003930701417544>

Tavares, António F., & Camões, Pedro J. (2010). New forms of local governance: A theoretical and empirical analysis of municipal corporations in Portugal. *Public Management Review*, 12(5), 587-608. <https://doi.org/10.1080/14719031003633193>

Tavares, António F., & Camões, Pedro J. (2023). Municipal Corporatisation in Portugal: From Mania to Depression. In Marieke Van Genugten, Bart Voorn, Rhys Andrews, Ulf Papenfuß, Harald Torsteinsen (eds.) *Corporatisation in Local Government: Context, Evidence and Perspectives from 19 Countries*, 315-333. https://doi.org/10.1007/978-3-031-09982-3_14

Voorn, Bart, van Genugten, Marieke L., & van Thiel, Sandra (2017). The efficiency and effectiveness of municipally owned corporations: a systematic review. *Local Government Studies*, 43(5), 820-841. <https://doi.org/10.1080/03003930.2017.1319360>

Williamson, Oliver E. (1979). Transaction-Cost Economics: The Governance of Contractual Relations. *Journal of Law and Economics*, 22(2), 233-261. <https://doi.org/10.1086/466942>

Williamson, Oliver E. (1981). The Economics of Organization: The Transaction Cost Approach. *American Journal of Sociology*, 87, 548-577. <https://doi.org/10.1086/227496>

